Practical Exercise: Static Analysis

Download **benign educational Windows executable** link
https://github.com/mpasco/MalbehavD-V1

Task 1 — File Identification

Download and Using Detect It Easy, record:

File type:
Architecture:
Compiler:
Entry point:
Number of sections:

Task 2 — PE Analysis

Using PE-bear, identify:

.text
.rdata
.data
.rsrc

and explain the purpose of each.

Task 3 — Strings

Extract **10 interesting strings** and classify them as:

- User-interface text
- File-related
- Network-related
- Configuration-related
- Other

Task 4 — Imports

Identify **10 imported functions** and explain what each function is normally used for.

Task 5 — Ghidra

Import the executable into Ghidra and identify:

1. Entry point
2. Main functions
3. Interesting strings
4. Important cross-references
5. Conditional branches

3.9 Review Questions

1. List the nine steps of basic static triage and state the analytical value of each.
2. Describe the major structures of the PE format in load order and explain the purpose of each.
3. Give five header-level indicators that a PE file is packed, and explain the reasoning behind each.
4. Explain the difference between ELF program headers and section headers, and why stripped static binaries are harder to analyse.
5. Explain how imphash and ssdeep differ from a SHA-256 hash and when each is preferable.
6. For each of the following capabilities — process injection, persistence, keylogging, ransomware encryption — name three indicative Windows imports.
7. Explain why a sample may import only LoadLibrary and GetProcAddress, and describe two more advanced variants of the same evasion.
8. Discuss the properties of a robust YARA rule and explain why hash-based detection alone is inadequate.